

would provide real-time data about the soil with the help of sensors to a cloud-based platform.

This cloud platform would also take information from satellites and suggest farming practices accordingly after proper data interpretation. The system also aims at giving the farmers – market information, post harvest advices and value addition options. This system in the far future also aims at removing the labour issues by setting up robotic farm equipments like sensor-based sprinklers that would do the farming practices that a labour generally does.



Distributed Robotic Garden

In many major countries also, Precision farming has gained a lot of traction. In Holland, presently Driverless tractors are being developed using Real Time Kinematic and GPS. It is proven to be quite effective and cost efficient for large farmlands

Researchers at MIT have setup a Distributed Robot Garden, which is a garden consisting of Tomato plants that are nurtured by Robots. The robots here water the plants, provide nutrients regularly studying the plants condition and harvest the tomatoes optimally. Each plant is equipped with sensors that update the robots about the plant's status; the whole garden has sensors that gives the map and respective positions of the plants to the robots and these robots move around and act according to the plant's condition. Robots right now are able to predict the state of the fruit, as in, when it would ripe and be ready for harvest and when the plant would require the next round of nutrients. Presently, the garden is open for students to conduct research and make this system better and make it commercially usable by Farmers.

Embedded systems can be also used to manage cattle in a better way. For ages now, managing cattle has been done with the help of fences that bind the cattle in a fixed area. However, this is very restrictive and these are physical fences and generally require a lot of money to be put up. Dr. Anderson (U.S. Dept. of Agriculture) came up with an idea of having virtual fences, instead of actually setting up real ones. This can be pos-